

DHRUV PRAJAPATI

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4th-Year Mechanical Engineering student experienced in hardware validation, thermal systems, and failure analysis across energy, manufacturing, and battery technologies.

EDUCATION

Rochester Institute of technology

Bachelor of Science in Mechanical Engineering, Senior Year, Dean's List 2026, GPA 3.50

Rochester, NY

Aug. 2023 – Dec 2027

TECHNICAL SKILLS

Thermal & Fluid Systems: Heat Transfer, Fluid Mechanics, CFD (SolidWorks Flow Simulation, ANSYS Fluent)

CAD Design & Analysis: SolidWorks, Ansys Mechanical & Fluent, CATIA v5

Programming: Python, MATLAB, Simulink, LabVIEW, Arduino

Hardware & Testing: Hitachi SEM, Oscilloscopes, Keysight Waveform generator, PIV, Instron

Manufacturing: CNC Milling, Lathe, GD&T, 3D Printing, Surface Metrology

EXPERIENCE

Battery Cell Engineer Intern

Tesla

Aug. 2026 - Dec. 2026

Austin, TX

- Incoming intern working with 4680 tabless cell mechanical, manufacturing and process development teams to support development at all stages.

Failure Analysis Engineer Intern

Pacific Gas & Electric Company, PG&E - ATS

Jun 2026 – Aug 2026

San Ramon, CA

- **Numerical Analysis of Substation Post Insulators** | *ANSYS Fluent + Maxwell + Structural, SolidWorks*
- Developed coupled CFD (transient), electromagnetic, and structural simulations using ANSYS to evaluate 4 operational loading conditions and identify critical stress concentrations in substation post-insulator assemblies.
- Supported validation of 80 in tall L-type 230 kV Lapp porcelain insulators through mechanical testing, data analysis, and technical reporting used for engineering assessment.
- **Materials Characterization** | *SEM, Microscopy, MATLAB, Root Cause Analysis*
- Conducted microscopy investigations on more than 5 failed utility assets, including switches, conductors, and post insulators, to identify fracture mechanisms and support root-cause analysis.
- **Residual Moisture Drying model** | *MATLAB, Experimental test*
- Developed a transient MATLAB evaporation model to predict residual moisture removal across a 14.7–1 psia pressure range, performing sensitivity studies on temperature and relative humidity effects after hydrostatic pipeline testing.

Mechanical Test Engineer Intern

MKS Instruments

Aug. 2025 – Dec. 2025

Rochester, NY

- **Liquid-cooled Heat Sink Analysis** | *SolidWorks Flow Simulation, experimental test bench*
- Designed and executed experimental characterization of a liquid-cooled microchannel heat sink for a 5 kW pulsed-DC generator (RPG-50), evaluating pressure-drop and flow-rate tradeoffs.
- Validated results using CFD simulations and first-principles analysis, contributing to a 25% improvement in system efficiency.
- **Cross-functional projects** | *ALT, Engineering Analysis, LabVIEW*
- Supported instrumentation and validation activities across multiple reliability test environments, including HVAC controls, liquid-nitrogen systems, and ALT/HASS/ESS chambers used for hardware qualification.

Mechanical Engineer Researcher

Fluids Acoustics Structures Laboratory, FAST LAB

Feb. 2026 – May 2026

Rochester, NY

- Designed and built an experimental vortex-ring test platform integrating a linear actuator, Arduino-based control system, and Pololu motor controller to investigate confined-flow fluid dynamics.
- Conducted Particle Image Velocimetry (PIV) experiments to quantify vortex-induced flow structures and impingement heat-transfer behavior in confined geometries.